Computer Vision and the Illustrated Book (Workshop)

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This workshop will introduce several fundamental techniques for the computational analysis of early European printed books (15th-19th centuries). Participants will gain a practical and theoretical understanding of the state of the art in computer vision applied to the illustrated printed book. They will learn how to make image collections searchable by means of a modular image processing pipeline composed of free and open-source tools, permitting illustration detection; full-page segmentation; illustration matching and classification; OCR; and document understanding. Materials will include early printed editions of Dante; Spanish and Scottish chapbooks; and nineteenth-century books and periodicals. Participants will learn how to apply, integrate and extend the software tools and processing pipeline to their own images; how visual search and analysis can scale to many millions of images; and will learn how computer vision can provide a deeper understanding of the visual content of image collections. Both the tools and the datasets are based on real-world research projects involving Oxford's Visual Geometry Group and collaborators in the digital humanities and cultural heritage fields.